

Michael Peirce

Senior Software Engineer

✉ michaelpeirce74@gmail.com ☎ +1 213 260 3999 📍 Morrow, OH

Profile

Senior Software Engineer with 11+ years of experience in **architecting** and **developing** scalable, high-performance applications using **Java**, **Spring Boot**, and **cloud technologies**. Passionate about writing clean, efficient code that adheres to industry-leading standards and continuously refining development practices. Experienced in designing **microservices architectures**, optimizing **databases**, and building secure, event-driven systems with **Kafka**, **AWS**, and **Azure**. Skilled in **full-stack development** with **React**, **Angular**, and **TypeScript** to create dynamic, user-friendly applications. Dedicated to **automation**, **DevOps** best practices, and hands-on with **Kubernetes**, **Docker**, and **CI/CD pipelines**. A proactive problem-solver who thrives in fast-paced, Agile environments, driving technical excellence and mentoring others.

Education

Bachelor's Degree in Computer Science

University Of Tennessee

2010 – 2013 | Knoxville, TN

Professional Experience

Senior Software Engineer

ITransition Group

02/2021 – Present | Decatur, GA

- Developed a **telemedicine-ready EMR/EHR system** using **Java** and **Spring Boot** for backend services. Leveraged a **microservices architecture** on **AWS** to enhance system interoperability, performance, and cloud efficiency.
- Built and optimized **RESTful APIs** with **Spring Boot**, ensuring seamless integration with third-party healthcare services (**FHIR**, **HL7 standards**) for patient record exchange and compliance with regulatory standards.
- Developed and maintained **cloud-native microservices** using **Spring Cloud**, implementing **Netflix Eureka** for service discovery and **Spring Cloud LoadBalancer** for efficient load balancing, ensuring high availability and seamless communication between microservices.
- Implemented robust **database management** using **Amazon RDS (PostgreSQL)** and **AWS DynamoDB**, optimizing data storage and query performance for high-volume patient records.
- Optimized complex **SQL queries**, implemented **indexing strategies**, and implemented **stored procedures** to improve database performance and response times in **PostgreSQL**.
- Integrated **Hibernate ORM** with **Spring Data JPA**, streamlining database transactions and improving efficiency in healthcare data processing.
- Implemented **serverless functions** using **AWS Lambda**, handling **event-driven** workflows such as automated appointment reminders, real-time patient data processing, and secure document uploads, reducing backend load and improving system scalability.
- Built an **event-driven** system with **Kafka**, **AWS SNS/SQS** for real-time patient data streaming, improving system responsiveness.
- Developed **AI-powered** clinical decision support and automated patient triage features using **Python-based machine learning** models (**TensorFlow**, **Scikit-learn**), improving diagnostic efficiency and provider workflows.
- Developed and optimized **ETL pipelines** using **Apache Spark** and **Apache Airflow**, enabling efficient extraction, transformation, and loading of large-scale healthcare data from multiple sources into a centralized data lake.

- Developed an **AI-powered medical Q&A system** with a **Node.js (Express.js)** backend (**OpenAI** integration using **openai-node** library), **MySQL** for database, **Vue.js** frontend, and **AWS** infrastructure, enabling automated responses for visitors while ensuring scalability, security, and fine-tuning capabilities.
- Designed and implemented high-performance **microservices** using **FastAPI (Python)** for specialized healthcare services, enabling efficient, low-latency APIs to complement the **Spring Boot**-based platform
- Developed a **patient portal** using **React.js**, implementing **Redux** for state management and integrating with backend services. The portal allowed patients to view their medical records, schedule appointments, and access telehealth services, improving engagement and accessibility.
- Built an **administrative dashboard** using **Angular**, leveraging **Angular Material** for UI components and **NgRx** for state management. This dashboard enabled healthcare providers to track patient progress, manage appointments, and streamline daily operations.
- Implemented real-time data visualizations and dashboards for clinicians using **D3.js** and **Chart.js**, enabling improved decision-making and clinical workflow efficiency.
- Integrated **real-time telehealth features** using **WebRTC** and **Twilio APIs**, facilitating secure video consultations and increasing system adoption for remote healthcare services.
- Managed **cloud infrastructure** on **AWS**, utilizing various services including **Amazon S3**, **AWS Lambda**, **Amazon API Gateway**, **Amazon CloudFront**.
- Developed automated **CI/CD pipelines** using **Jenkins**, **GitHub Actions**, and **Terraform (Infrastructure as Code - IaC)**, ensuring efficient deployments and minimizing errors.
- Integrated **SonarQube** for static code analysis, identifying vulnerabilities, enforcing best practices, and ensuring code security and maintainability.
- Followed **Test-Driven Development (TDD)** with **JUnit**, **Mockito**, and **TestNG** for backend **unit** and **integration testing**. Implemented **end-to-end UI** tests using **Cypress** and **Selenium**.
- Engineered secure authentication and authorization mechanisms using **OAuth2**, **JWT**, and **Spring Security**, ensuring compliance with **HIPAA** and **GDPR** regulations.
- Enhanced logging, monitoring, and performance analytics using **Splunk**, **Prometheus**, and **Grafana**, enabling real-time observability, log analysis, and proactive troubleshooting for healthcare applications.
- Led **Agile Scrum** teams, **mentoring** junior developers and conducting **code reviews**, improving team efficiency and development velocity.

Senior Software Engineer

Thoughtworks

10/2017 – 01/2021 | Chicago, IL

- Designed and developed microservices using **Spring Boot** for **large-scale eCommerce** and **enterprise applications**, including product catalog management, order fulfillment, and dynamic pricing optimization.
- Built scalable **RESTful APIs** using **Spring Boot**, **Spring Data JPA**, and **Hibernate**, ensuring efficient data transactions and seamless communication between microservices and external systems.
- Developed and optimized a large-scale product catalog system, integrating supplier data via **ETL pipelines** and managing structured storage in **Oracle Database** with **Hibernate ORM** for efficient querying.
- Implemented **Azure Functions** for critical backend workflows, including order confirmation processing, tax calculation, and fraud detection, enhancing transaction reliability and compliance.
- Implemented real-time dynamic pricing models leveraging **Kafka**-driven price adjustments, enabling automated competitive pricing strategies based on demand fluctuations, supplier costs, and competitor analysis.
- Built an intelligent supply chain forecasting engine using **Apache Spark**, optimizing inventory levels, demand predictions, and warehouse logistics to reduce costs and improve stock availability.
- Utilized **MongoDB** for storing and managing semi-structured data, improving data retrieval efficiency and scalability in high-performance applications.
- Integrated **Azure Cache for Redis** for frequently accessed product details and high-traffic API responses, reducing database load and improving performance.

- Developed an internal analytics and reporting tool using **Django(Python)** and **Angular**, integrating with **Spring Boot microservices** to provide real-time insights into system performance, customer behavior, and operational efficiency.
- Worked extensively on **front-end development**, coupled with full front-end work (**HTML, CSS, JavaScript, TypeScript, Angular, React**), ensuring a seamless, high-performance user experience.
- Heavily used **Microsoft Azure Services**, integrating **Azure Functions, Azure Service Bus, Azure Storage**, and more to build scalable, **event-driven microservices**.
- Deployed **microservices** on **Kubernetes** with **Docker**, ensuring automated, containerized deployments across multiple cloud environments.
- Utilized **Azure DevOps** for **CI/CD pipelines**, automated deployments, and version control, ensuring seamless integration, **continuous delivery**, and efficient collaboration across development teams.
- Integrated **Azure Service Bus** for asynchronous messaging, ensuring scalable, reliable processing of supplier updates, order fulfillment, and returns handling.
- Implemented **OAuth2** and **JWT**-based authentication to secure API endpoints, ensuring robust access control and compliance with industry security standards (**GDPR, PCI-DSS**).
- Integrated third-party payment APIs (**Stripe, PayPal, Apple Pay**) into backend microservices, ensuring secure transaction processing and compliance with **PCI-DSS** standards.
- Created custom monitoring dashboards in **Kibana (ELK Stack)** to track real-time application health, logs, and error management, improving observability and proactive issue resolution.
- Ensured code quality and application reliability by incorporating **Test-Driven Development (TDD)** principles, reducing defect rates and improving maintainability.
- Developed and automated testing strategies using **JUnit, TestNG, Mockito** for backend services, and **Jest, Selenium** for front-end UI testing, increasing test coverage and ensuring application reliability.
- Actively collaborated in **Agile Scrum** teams, participating in sprint planning, code reviews, and mentoring junior developers, contributing to faster release cycles and enhanced platform stability.

Software Developer

Elite Business Solutions

08/2013 – 09/2017 | North Royalton, OH

- Developed a web application using **Java, Spring MVC, JSP, Thymeleaf**, and **Servlets**, handling invoice generation and payment tracking.
- Designed the relational database using **MSSQL** and **JDBC**, implementing complex **SQL** queries to retrieve customer billing records.
- Implemented session-based authentication using **Java Servlet Filters**, securing access to customer financial data.
- Generated PDF invoices using Apache POI and iText, allowing clients to download statements and receipts.
- Developed a multi-tier HRMS system using **JSF (JavaServer Faces)** and **PrimeFaces**, providing a component-based UI for HR staff to manage records efficiently.
- Implemented CRUD operations using **Hibernate ORM**, enabling seamless employee record retrieval, updates, and payroll calculations.
- Built a scheduled batch job using **Spring Batch**, automatically processing monthly salary disbursements.
- Integrated **JavaMail API**, allowing automated notifications for leave approvals and HR policy updates.
- Designed and developed a modern Employee Self-Service (ESS) portal using **AngularJS**, providing real-time updates and a dynamic user experience.
- Developed backend components using **Java/J2EE** technologies, applying **Core Java, Multithreading, Java Collections, Java I/O**, and **JDBC** to build efficient and reliable business logic.
- Exposed **RESTful APIs** with **Spring Boot** and **Spring Data JPA**, allowing seamless data exchange between the frontend and backend.
- Migrated file storage to **AWS S3**, ensuring scalable and secure document management for employee records.

- Implemented authentication using **OAuth2** and **JWT**, replacing the old session-based authentication model.
- Deployed the application on **AWS EC2**, utilizing **AWS RDS** for database hosting and **AWS CloudFront** for asset delivery.
- Developed a **microservices-based** customer support system using **Spring Boot**, integrating **RabbitMQ** for event-driven ticket processing, ensuring scalable and real-time updates for customer support interactions.
- Built a ticket management **UI** using **AngularJS**, allowing customers to raise and track support tickets in real time.

Skills

Programming Languages & Frameworks

- Backend: Java (Spring Boot, Spring MVC, Spring Security, Spring Cloud, JPA/Hibernate), Kotlin, Python (FastAPI, Django), Node.js (Express.js), C/C++ (system-level programming, memory management), C# (.NET ecosystem)
- Frontend: React.js, AngularJS, Angular(8, 13, 16), Vue.js, jQuery, TypeScript, JavaScript (ES6+), HTML, CSS, Bootstrap, Material UI, Thymeleaf

Cloud & DevOps

- AWS (EC2, Lambda, EKS, Fargate, S3, RDS, DynamoDB, ElastiCache, VPC, API Gateway, WAF, IAM, Cognito, CloudFront, Route 53, SNS, SQS, Kafka (MSK), EventBridge, Step Functions, CloudFormation, CDK, Terraform, CloudWatch, AWS X-Ray, Security Hub)
- Azure (Virtual Machines, AKS, Azure Functions, Azure SQL Database, CosmosDB, Blob Storage, Azure Cache for Redis, Azure AD, API Management, Key Vault, App Gateway, Service Bus, Event Grid, Kafka on Azure, Azure DevOps, Terraform, Bicep, Application Insights, Log Analytics, Azure Monitor)
- GCP

Databases & Data Management

- SQL Databases: PostgreSQL, Oracle, MySQL, Microsoft SQL Server, Amazon Aurora, PL/SQL
- NoSQL & Caching: MongoDB, DynamoDB, Neo4j, Redis, Elasticsearch, Cassandra
- Big Data & ETL: Apache Spark, Apache Airflow, AWS Glue, Athena, Azure Data Factory
- Event-Driven Architectures: Kafka, RabbitMQ, ActiveMQ, Azure Event Hubs

APIs & Communication Protocols

- RESTful APIs, SOAP, GraphQL, gRPC
- JSON, XML

Software Architecture & Design

- Microservices, Event-Driven Architecture, Serverless Computing, Domain-Driven Design (DDD), API Gateway, CQRS, SOLID Principles

Security & Compliance

- Identity & Authentication: OAuth2, JWT, Spring Security, Keycloak, OpenID Connect
- Cloud Security & Compliance: AWS IAM, Security Hub, WAF, Azure AD, Key Vault
- Regulatory Compliance: HIPAA, GDPR, PCI-DSS

Testing & Quality Assurance

- Backend: JUnit, TestNG, Mockito, Cypress, Selenium, Mocha
- Static Code Analysis & Monitoring: SonarQube, Prometheus, Grafana, Splunk, Kibana

Development Methodologies & Tools

- SDLC, Agile (Scrum, Kanban), Jira, Confluence, Git (GitHub, GitLab, Bitbucket)
- CI/CD Pipelines: Jenkins, GitHub Actions, GitLab CI/CD, Azure DevOps, AWS CodePipeline
- Infrastructure as Code (IaC): Terraform, CloudFormation, Bicep, CDK
- Code Reviews, Pair Programming, Mentoring